

[Page 1]

Application of Digital Twin Technology

Digital Twin - Cyber Security Simulator

Digital Twin Solutions

Reality Capture

3D Laser Scanning Photogrammetry 360°

Videogrammetry

3D As-Built Models for Digital Twin

Build Low Carbon Smart Plant with Indus -
trial Digital Twin

Information Engineering for Digital Twin

The Leader in Industrial Telecommunication,
Industrial IT, Industrial Network and Security

Empowering Digital Transformation

Digital Twin &
Solutions

READ MORE

[Page 2]

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A Digital Twin is a virtual representation of a physical asset, system, or process. It combines real-time data, 3D models,

and simulations to create a dynamic digital counterpart that reflects the actual state and behaviour of the physical entity.

Benefits of Digital Twin Technology:

Enhanced Visualization: Offers a 3D view of assets, making it easier to understand complex systems.

Real-time Monitoring: Integrates IoT data for continuous tracking of performance and health.

Predictive Maintenance: Analyzes data to forecast potential failures, reducing downtime and maintenance costs.

Improved Design and Engineering: Facilitates better design choices through simulation and scenario analysis.

Operational Efficiency: Optimizes processes and resource management through data-driven insights.

Industries We Serve

Oil & Gas Petrochemical Power Buildings &

InfrastructureVector InfoTech seamlessly integrates Digital Twin creation capabilities with proven software solutions, empowering

engineers, architects, and facility owners to effectively leverage this technology for engineering, design, operations, maintenance, and asset management initiatives.

[Page 6]

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Build Low Carbon Smart Plant with Industrial
Digital Twin
Industrial Digital Twins Plant Solution Architecture
Emission source settings
Carbon emissions inventory and benchmarking
Emission reduction plan design
Carbon management plan executionIndustrial Digital Twin Plant
Equipment operation and maintenance
Static Equipment Corrosion Management
Visualized intelligent inspection
Production process optimization
Intelligent management and control of security risks
Multiple coupling risk assessment
Safety training and simulation drills

[Page 7]

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Information Engineering for Digital Twin

Through Reality Capture and 3D modelling, the visual and physical representation of an asset can be captured and presented digitally in a software to represent part of its Digital Twin DNA. To complete the Digital Twin, static and live data of the object need to be captured, structured and contextualized as information related to the object.

To create a Digital Twin of an industrial plant, the following steps are required. Vector InfoTech provides the following consultancy & services to:-

1. Define the scope and objectives of your Digital Twin project.
2. Choose the modelling and simulation tools that suit your needs and capabilities.
3. Develop and validate the Digital Twin model.
4. Integrate the Digital Twin with your industrial control system.
5. Monitor and maintain the Digital Twin.

effectively leverage this technology for engineering, design, operations, maintenance, and asset management initiatives. Our approach enhances collaboration and decision-making, enabling organizations to optimize performance and efficiency throughout the life-cycle of their assets. With our expertise, clients can confidently navigate the complexities of modern engineering and facility management.

We specialize in creation and utilization Visual, 3-Dimensional and Information Digital Twin for the Oil & Gas, Petrochemical, Power industries and for Building & Infrastructures.

Monitoring production
processes, pipeline integrity,
and asset management.Streamlining operations and
enhancing safety through

real-time data.Optimizing energy
production and distribution
systems.Managing facilities efficiently
for improved occupant
experience and safety.

[Page 3]

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Application of Digital Twin Technology

Digital Twin technology is now being adopted in the complete life cycle management of Asset Intensive industries.

Vector InfoTechs Digital Twin & solutions applications span various industries and functions, providing significant benefits in efficiency, performance, and decision-making. Heres a look at some key applications across different sectors:

Digital Twin - Cyber Security Simulator

Asset Management

Production Management

Environmental Management

Energy Carbon Management

Planning

Engineering Design

Architectural Design

Construction

Operation & Maintenance

Asset Performance

Facility Management Module

Digital Twin - Internet of Things (IoT)

Digital Twin - RTLS (Real Time Location and Assets Tracking)

1. Digital Twin Creation: Tailored services to develop 3D models and simulations specific to your assets.

2.

4. Integrate the Digital Twin with your industrial control system.

5. Monitor and maintain the Digital Twin.

In the life cycle of a facility, a multitude of engineering information is created, utilized, changed and stored in multiple applications. These information in hard copy or in digital format is the engineering representation of the actual facility. Changes to the engineering information and the physical plant occurs periodically in the life-cycle of a facilities. The above scenarios compounds the difficulties in the (continuous) successful implementation of an intelligent engineering and data management initiative. Vector FSD specializes in capturing, consolidating and standardizing 1D and 2D engineering information into digital software solution as part of the Engineering Digital Twin.

P&ID

Line List

Instrument List

Equipment List

Valve List/MTO

SP Item List

P&ID legend sheet

P&ID Drawing

P&ID Instrumentation

Instrument Index

IO list

Alarm Trip Setting List

Datasheet/Process datasheet

Calculation Sheet

Cable and Panel Schedule

MTO

Wiring Termination Drawing

Loop Drawing

Hook up/Installation DrawingElectrical

Equipment List

Load List

Cable Schedule

Small Power/Lighting/UPS

Panel Schedules

Single Line Diagram

Switchgear Schematics

Motor Schematics

Connection DiagramWith our pool of personnel, we can provide support for system implementation, administration and outsourcing.

Application configuration & Set-Up

Data Loading

Administrator and User Training

Development of User Guides and Procedures

Specializing in intelligent engineering & design applications (2D & 3D)With the ever-growing implementation of Digital Twin technology utilizing advanced software applications, it is vital that organizations implementing such solutions have the support for correct setup and administration of these intelligent tools and creating & maintaining the data.

Things (IoT)

Digital Twin - RTLS (Real Time Location and Assets Tracking)

1. Digital Twin Creation: Tailored services to develop 3D models and simulations specific to your assets.
2. Data Integration: Seamless integration of IoT and operational data for real-time monitoring and analysis.
3. Visualization Tools: Advanced software solutions that provide interactive visualizations for better

understanding and decision-making.

4. Predictive Analytics: Utilizing machine learning algorithms to predict trends and maintenance needs.

5. Training and Support: Comprehensive training programs and ongoing support to ensure effective implementation and use.

By leveraging our Digital Twin solutions, organizations can achieve greater operational resilience and make informed strategic decisions. Let us help you unlock the full potential of your assets!

Digital Twin Solutions

Digital Twin - Cyber Security Simulator

To enhance and assess the skills of IT engineers and operators, a Digital Twin of the Cyber Security ecosystem of the actual live plant is replicated. Incorporating a mock-up of the plant can significantly enhance the realism of training and testing environments for operational technology systems.

This can be achieved by way of the OT Scada Cyber Range which contains elements of the actual control and shutdown system together with the actual cyber security architecture.

[Page 4]

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3D Laser Scanning Photogrammetry 360°

Videogrammetry

Vector FSDs Reality Capture methodology is centered around our core service of 3D Laser Scanning and Photogrammetry. Our advance workflow enables us to provide the fastest, the most complete and most accurate Reality Capture service.

tions have the support for correct setup and administration of these intelligent tools and creating & maintaining the data. This is a pre-requisite for the successful implementation of any Digital Twin initiative for the end user to obtain on going return on the investment.

[Page 8]

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www.vectorinfotech.com Vector InfoTech Group

Vector InfoTech Group (Vector InfoTech) is a Singapore grown global engineering group specializing in AI & Data Hub, Digital Twin, Digital Infrastructure, Cyber Security and Smart Energy & Sustainable Technologies solutions.

Incorporated in 1992, the Group today operates in over

13 countries, has a sizable and growing workforce of over 300 employees, serves both commercial and industrial sectors across the globe, and offers a unique blend of products/services of highly advanced industrial, smart and sustainable business solutions.

The Groups business focuses on three main streams:

Distribution, Turnkey Integrated Project Solutions, and Maintenance Services. It has a reputable presence and firm foothold across various industries, attributable to the groups large spread of quality clientele comprising of global multinationals and state-owned enterprises with business interests in essential infrastructures, oil & gas, petrochemicals, power & utilities, transportation, chemicals, building & construction, renewable energies, Food, Pulp & Paper Industries.

We specialize in providing comprehensive system solutions that encompass Industrial Communications & Networking Systems Hard -

ware, Software and Engineering Services. Our expertise extends to delivering Total Turnkey Communication System Packages.

Our advance workflow enables us to provide the fastest, the most complete and most accurate Reality Capture service.

Together with our technology partners, Vector FSD is pioneering the use of 3D Point Cloud and 3D Mesh model as the Visual-Dimensional Digital Twin.

3D Point Cloud captured with Laser ScanningFor creation of Visual-Dimensional Digital Twin, Vector FSD utilizes Reality Capture technology. Reality Capture

combines specialized high technology equipment and software to capture data of the physical asset or environment. The data captured is then processed and converted into 3D digital representation of the physical asset in form of high-resolution Point Cloud, Reality Mesh model or 3D CAD model.

Reality Capture

[Page 5]

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3D As-Built Models for Digital Twin

Vector FSD Digital specializes in conversion of Industrial Plant, Building, Infrastructure and Marine reality capture dataset into As-Built 3D CAD Model. The 3D CAD model is converted and delivered in industry standard software solutions as a 3D Primitive model, 3D Semi-Intelligent Tagged model or a 3D Intelligent model where model objects are created based on Specifications & Catalogues. The 3D Intelligent model forms the Engineering Digital Twin becomes the base information for development of other Digital Twin application.

Vector FSD Digital has implemented and delivered As-built 3D model for implementation of Digital Twin technology in Brownfield Engineering & Design Plant Revamp projects and Asset Information Management (AIMS) projects.

To convert 3D laser scan point clouds into 3D CAD model, we utilize the latest software solution in the market.

3D As-Built Models for

Industrial Plant Digital Twin3D As-Built BIM model for

Building Digital Twin

ervices. Our expertise extends to delivering Total Turnkey Communication System Packages.

Our proven track record demonstrates our success in delivering reliable, high-performance products and complete solution systems. We

have established consistent, cordial, and strong working relationships with renowned local and global organizations from diverse indus -

tries, further solidifying our achievements.

Vector InfoTech Expertise

MALAYSIA

SarawakSAUDI ARABIA

Dammam

SINGAPORE

(HQ) U.A.E

DubaiINDONESIA

JakartaPHILIPPINES

ManilaSOUTH KOREA

Seoul

THAILAND

Bangkok

VIETNAM

Ho Chi MinhJAPAN

TokyoINDIA

Bangalore

Chennai

MALAYSIA

Kuala LumpurBAHRAINBANGLADESH

DIGITAL TWIN

& SOLUTIONS

Vector InfoTech delivers

innovative digital tech -

nology solutions that

harness modern software,

cloud, AI, and IoT, ana -

lytical modelling for big

data, for a competitive

edge in digital transfor -

mation. Our offerings,

including Digital Twin,

IoT platforms, and Asset

Management systems,

are customized for Smart

Cities, Smart Plants, Smart
Vessels, and Connected
Workplace Management,
among other industries.DIGITAL
INFRASTRUCTURE

Using our deep exper -
tise in LAN, Fiber Optic,
Video, Data, 3G/4G/5G,
LoRa, types of commu -
nication technologies,
Vector InfoTech provides
reliable and powerful yet
economically viable solu -
tions for Digitalization
across all industry verti -
cals, including critical &
essential infrastructures
such as airports, industri -
al plants, pipelines, trans -
portations, etc.

CYBER

SECURITY

Vector InfoTech provides
superior Cyber Security
solutions for the most
critical infrastructures.

We are highly trusted by
many enterprises inter -
nationally to provide net -
work security and cyber
security solutions and
services for both the IT OT
systems.SMART ENERGY &

SUSTAINABLE

TECHNOLOGIES

With the adoption of
COP26, global demand
for de carbonizing and
greener infrastructures
are increasing.

ES

With the adoption of

COP26, global demand for de carbonizing and greener infrastructures are increasing. Vector InfoTech has incorporated digital solutions with Energy Technology (ET) to harvest big data to provide de-carbonization capabilities to businesses.

Scan for contact details

Scan for brochures

AI & DATA HUB

SOLUTIONS

AI & Data Hub Solutions integrate advanced analytics and centralized data management to enable organizations to unlock insights, automate processes, and enhance collaboration. By leveraging AI's predictive capabilities and the comprehensive access provided by data hubs, businesses can make informed decisions, improve customer experiences, and drive innovation in a rapidly changing environment.